

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF
ECONOMICS Ordinary Linear Regression

Week 12 · Least Squares, Residuals and L-I-N-E · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 Academic
Year



From “how strong” to “how”

What we've done so far

Week 6: r — strength and direction of the relationship. Week 11: Cramer's V — strength of the categorical relationship.

Both are the same number. Both answer the question “how much”.

$r = 0.83$ does not answer either of these questions.

Regression — the SHAPE of the relationship

Instead of r , there are two numbers: the slope and the intercept. Together, they give a line — that is, a specific answer for each x .

And with the line comes something new: residuals. Half of this week is about residuals — because they are the only thing that tells you where the model doesn't work.

A common mistake

No inference this week. The t -test for slope, confidence intervals, and prediction intervals are in week 13. Today is just: constructing a line, reading it, and whether you can trust it.

Lecture 1 — Building a Line 1. Least Squares: What is Minimized and Why?

2. Manually Calculating β_1 and β_0 on Real Data; The Relationship $\beta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)}$.

3. The Residual and Its Two Identity.

4. What R^2 Means — and Why R^2 Is More Useful Than It.

Lecture 2 — Believing in a Line 5. The L-I-N-E Conditions, Each Violated in a Real-World Example.

6. Leverage, Residual, and Cook's Distance — Three Other Concepts.

7. Anscombe's Quartet (from Week 6), Now with Residuals.

8. Regression to the Mean — measured.

9. Extrapolation: where the model gives a meaningless response.

Conditions CLO8. Describe the relationship between two variables with a linear model, interpret the results, and test the model's conditions and state its limits. Bloom: Apply → Analyze → Evaluate.

Least squares

What is minimized — and why?

Model (F159)

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

β — prime set (unknown), $\hat{\beta}$ — sample estimate. The difference between a Greek letter and a Latin letter has not changed since week 9.

Criterion (F160)

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

Solution (F161–F163):

$$b_1 = \frac{S_{xy}}{S_{xx}}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

F162 — Bridge to week 6

$\rho = \frac{S_{xy}}{S_x S_y}$ Slope — the unit form of correlation.

Why squares? Three reasons, all three are valid

1. The square loses sign — positive and negative deviations do not cancel each other out.
2. The solution is in closed form: setting the derivative to zero gives two linear equations. There is no such formula for the absolute value.
3. If ε is normal, least squares overlaps with the maximum confidence estimate (week 9).

And a confession: The square gives too much weight to large deviations—so least squares is not stable for outliers (we will see this in section E). Minimizing the absolute value is more stable, but it is computationally difficult. This choice is for convenience, and should be stated as such.

Real data, calculated manually

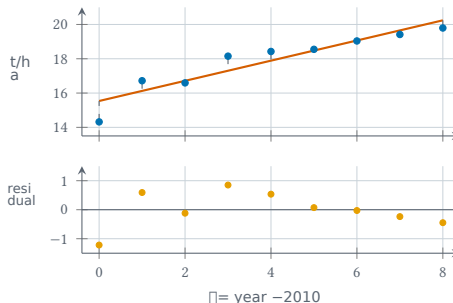
Kazakhstan, potato, 2010-2018

x = year - 2010, y = yield (t/ha), n = 9

year x y $\hat{y}$ e

2010	0	14,319	15,5373	-1,2183
2011	1	16,718	16,1253	+0,5927
2012	2	16,593	16,7134	-0,1204
2013	3	18,152	17,3014	+0,8506
2014	4	18,425	17,8894	+0,5356
2015	5	18,550	18,4775	+0,0725
2016	6	19,038	19,0655	-0,0275
2017	7	19,416	19,6535	-0,2375
2018	8	19,794	20,2416	-0,4476

Conditions $\sum x = 4$, $\sum y = 17.889444$, $\sum xy = 60$, $\sum x^2 = 35.282000$
 $\sum y^2 = 23.870096$, $\hat{y}_0 = 15.537311$, $\hat{y}_1 = 0.588033$, $\hat{y}_2 = 0.932289$



Model and its interpretation

$\hat{y} = 15.537311 + 0.588033 x$ $\sum (x - \bar{x})^2 = 60$ for $x = 0..8$.

$\hat{y}_1$: the average annual yield increased by 0.5880 t/ha. $\hat{y}_0$: the model value in 2010 was 15.5373 t/ha ($x = 0$ is meaningful here — not always!).

$\sum e = -1.78 \times 10^{-14}$ (F166), $\sum e^2 = -6.75 \times 10^{-14}$ (F167) — it is no coincidence, these are the least squares equations themselves. The line passes through $(\bar{x}, \bar{y})$ (F181).

□2 and □□

Dividing the deviation in half

Three sums (F168-F171)

$SST = \sum (y_i - \bar{y})^2$ total deviation

$SSR = \sum (\hat{y}_i - \bar{y})^2$ explained by the model

$SSE = \sum (y_i - \hat{y}_i)^2$ remainder

$SST = SSR + SSE$ — identity

In our example: $23.870096 = 20.746992 + 3.123104$; difference 7.11×10^{-15} .

R^2 (F172-F173)

$$R^2 = \frac{SSR}{SST} = 0.869162$$

In a simple regression, $R^2 = R^2$ (difference 1.11×10^{-16}). “86.9% of the variability in y is due to x .”

s_e (F174-F175) — and why it matters

$$s_e = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{3,123104}{7}} = 0,667950$$

Unit t/ga — same as $\bar{y}$. $s_e = 1.727357$; the model reduced the standard error by 61.33%.

$n-2$: two parameters were estimated.

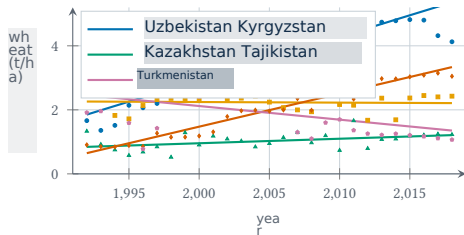
Two models, same R^2 , different accuracy

potatoes ~year R^2 s_e (t/ha)

Uzbekistan	0.969364	1.502218
Kazakhstan	0.901928	1.208462

The difference in s_e is only 0.067435, but s_e is 1.2431 times different — and it is s_e that gives the prediction error. R^2 — how much the model used x ; s_e — the prediction error. Only s_e answers the question “How accurately do I predict?” Always include both.

One crop, five countries: β_2 range from 0.00 to 0.93



country β_1 β_2 β_3

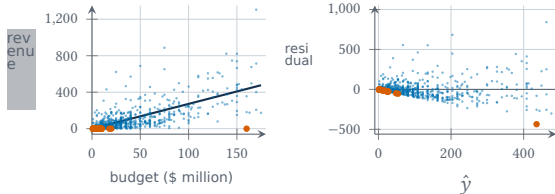
Tajikistan	0.10340	0.9262	0.2362
Uzbekistan	0.13486	0.8237	0.5051
Kazakhstan	0.01411	0.1857	0.2392
Turkmenistan	-0.04259	0.1750	0.7485
Kyrgyzstan	-0.00189	0.0029	0.2825

Three questions to the class

1. In Kyrgyzstan, $\beta_2 = 0.0029$. Does this mean “the model is bad” or “there is no growth”? (The latter — and this is a VALUABLE result.)
2. In Turkmenistan, $\beta_1 < 0$. Has productivity declined? (The model says so — but $\beta_2 = 0.175$, too noisy.)
3. In Tajikistan, β_2 is the highest. So is agriculture there the most stable? (No — it’s about β_2 TREND.)

Remains

A residual plot finds an ERROR in the data



Conditions

$\hat{\beta}_0 = -0.705390$ and $\hat{\beta}_1 = 2.726829$. The red dots are the straight line below.

What is the red line?

31 films have had a worldwide gross of exactly zero, and all 31 of those have had a domestic gross of zero. The biggest one is Aquaman (2018), budget \$160 million, gross of 0. That's impossible.

So zero is not “no income”, but “no data”. This was found by the residual plot, not a statistical test.

And now the honest part

We remove them: $\hat{\beta}_1$ 2.726829 $\rightarrow$ 2.739510 — a change of only 0.4650%. 31 out of 2,467 did not change the result much.

But the correction is still **CONDITIONAL** — next time $n = 30$, and then one such point will be reached (section E).

L-I-N-E: four conditions, four questions

Conditions and their verification tool

Condition Tool

L Linearity $\hat{\beta}$ vs $\hat{\epsilon}$ — no shape

I Independence $\hat{\epsilon}$ vs order; lag-1; Durbin-Watson

N Normality QQ-plot (not a test!)

E Equal variance $\hat{\epsilon}$ vs $\hat{\epsilon}^2$ — no fan

WHY are conditions needed — clear answer

No conditions are needed to calculate $\hat{\beta}_0$ and $\hat{\beta}_1$.
Least squares always gives a number.

Conditions are needed for $\hat{\beta}$, prediction, and the conclusion in week 13 ($\hat{\beta}$ -test, confidence interval). If the condition is violated, the slope is still calculated — it just can't be trusted.

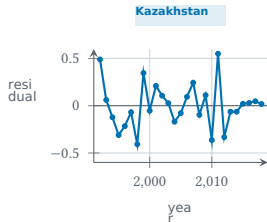
Order of importance — not equal

1. L — most important. If violated, the model answers the wrong question.
2. I — very important. If violated, $\hat{\beta}$ and all $\hat{\epsilon}$ -values are wrong (usually very small).
3. E — important. If violated, the estimate is reasonable, but the precision estimate is wrong.
4. N — least important. If n is large, the central limit theorem (week 7) makes it unnecessary.

A common mistake

Students usually do the opposite: they test for normality and don't check for linearity. The most important condition is the easiest to see — just look at the diagram.

Condition I: Week 11 warning returned



Uzbekistan Kazakhstan

ρ^2 0.8237 0.1857 lag-1 corr.
0.789704 -0.293276 limit 0.377195
0.377195 DW 0.387164 2.365583
condition VIOLATED fulfilled

What does it mean

Limit $\pm 0.975/\sqrt{27}$. Durbin-Watson (F180) is independent ≈ 2 ; theoretical $2(1 - \rho) = 0.420592$ — observed 0.387164.

In the Uzbekistan series, the positive deviation in one year is maintained in the next year. So 27 observations are not 27 independent data.

Conditions Same model (wheat ~ year), same years. Residuals on the left form waves; those on the right are random jumps.

Audience question

In week 11, this EXACT same series violated the independent condition — then in the two-sample test, today in the regression. This is the most repeated idea of the course: the condition is more important than the formula, and the same data defect recurs in each method.

Condition E: logarithm does not lose its fan shape — it inverts



Conditions

$$n = 2436 \cdot b_1 = 1,148411 \cdot R^2 = 0,466680 \cdot s_e = 0,695197$$

F183 - interpretation of the log-log model

$\epsilon_1 = 1,1484 \Rightarrow$ if the budget increases by 1 percent, revenue increases by 1.1484 percent on average (elasticity). $\epsilon_1 > 1$: revenue grows faster than the budget.

model residual SD, budget quarter ratio

raw	34.01	58.78	75.76	178.20	5.2395	log-log
	1.0196	0.7302	0.5156	0.3706	2.7513	

This was not the expected result

In the raw model, SD increases with budget (5.24 times). In the log model, SD decreases with budget (2.75 times).

That is, the logarithm did not lose its fan - it turned it upside down and almost halved its power. Condition E is violated in both models.

The reason is economic: a \$ 1 million film can collect both 0.1 million and 100 million - the relative result is very variable. A big-budget film is relatively more predictable.

Audience question

Rule: rotation is not a magic wand. It changes which three are noisy, it does not eliminate noise. The residual plot is redrawn after each transformation. And the ϵ_2 of the log model cannot be compared to that of the raw model—they explain different ϵ .

Touching points

Leverage, residual, Cook's distance — three other things

Three definitions

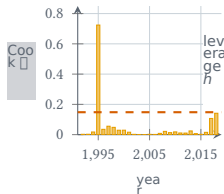
(E176-E178)

$\hat{y}_i$, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, $\text{stud}^2 = \frac{1}{n-2} \sum (y_i - \hat{y}_i)^2$ — how far along $\hat{y}$ (independent of $\hat{y}$) · $\hat{y}_i$ — how far from the line · $\hat{y}_i$ — how much the line without a point moves. The limit is $\hat{y}_i > 4/n$.

Only the third is “impact”

Large leverage + small residual $\Rightarrow$ no impact. Small leverage + large residual $\Rightarrow$ small impact. Large leverage + large residual $\Rightarrow$ risk. $\sum h_i = 2$ always; average $h = 2/n$.

Tajikistan, potatoes

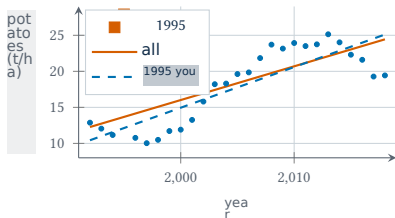


The two diagrams are completely different

Leverage is U-shaped: the outermost years are the largest — 1992, $h = 0.140212$, and this is determined without any regard to $\hat{y}$. The Cook distance explodes in one year: 1995, $\hat{y} = 0.724102$, the limit $4/27 = 0.148148$.

1992 has the largest leverage, but the Cook distance is only 0.002065. The 1995 year is $h = 0.098087$ — slightly above the average ($2/n = 0.074074$). What makes it impressive is not the leverage, but the studentized residual = 3.649147.

Two influential points, same diagnosis, TWO DIFFERENT correct answers



Conditions Tajikistan: 1994 = 11,167, 1995 = 27,900, 1996 = 10,770 t/ha. More than twice as high as its neighbors.

Tajikistan potato all 1995 you

Δ 1 (t/ha per year) 0.468036 0.564322 Δ 2
0.460020 0.712811 Δ 4.104564
2.863862

Δ 1 change +20.5723 %

Two points, two different decisions

Tajikistan 1995. Civil war period (1992-1997). No mechanism explains the double jump and the value falls back the following year. This is probably an error in the source. $\Rightarrow$ Check the source; correct if confirmed, otherwise remove and write it down.

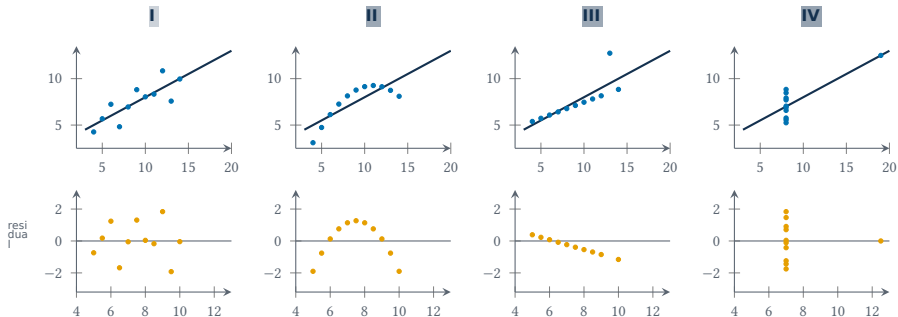
Kazakhstan rice 1992 ($h = 0.140212$, the largest leverage and the largest impact). 1992 = 3.992, then drops to 2.193 by 1995 and slowly recovers. This is real history: the collapse of the Soviet irrigation system. $\Rightarrow$ Save the point, change the model.

Audience question

Statistical ethics. Removing an outlier “to make the result look better” is a form of Δ -hacking in regression from Week 11. The rule is one: the decision comes from the data source, not the result — and both options are reported.

Anscombe and reversion to the mean

Anscombe Quartet — from week 6, now with residuals



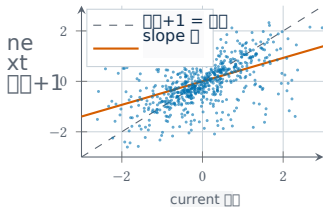
	b_0	b_1	R^2	s_e
I	3.0001	0.5001	0.6665	1.2366
II	3.0009	0.5000	0.6662	1.2372
III	3.0025	0.4997	0.6663	1.2363
IV	3.0017	0.4999	0.6667	1.2357

σ_1 range 0.0004 · σ_2 range 0.0005

IV — limit state

$h=19$ The leverage of the point $h=1.0000$ is exactly 1; the remainder is -1.78×10^{-15} is exactly 0. The line is forced through it, so the Cook distance is undefined (0/0) — not “large,” but meaningless. Without this point, $\hat{\beta}_1=0$; $\hat{\beta}_1$ does not change in the remaining 10 observations, so the slope cannot be calculated. All $\hat{\beta}_1$ comes from a single observation — $\hat{\beta}_2=0.6667$ does not indicate this.

Reversion to the mean: measured, not stated



Conditions 25 rows (5 countries $\times$ 5 crops), residuals from the trend standardized. $n = 650$ pairs.

Galton (1886) — and why this is not biology

$\rho = 0.464954$ — positive. That is, a good year is expected to be followed by another good year.

However, extreme values revert to the mean:

group current next

$z > 1$	96	+1,5817	+0,6719
$z < -1$	83	-1,5269	-0,7200

Theory tells the number IN ADVANCE (F182)

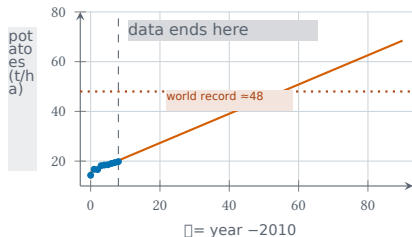
$E(\rho + 1 | \rho) = \rho = 0.464954 \times 1.5817 = 0.7354$ Observed: 0.6719. Difference 0.0635.

Reversion to the mean does not mean that “the top group got worse”. It is simply a consequence of $\rho < 1$. There is no reason for it.

A common mistake

A false “effectiveness” conclusion. If a “help program” is applied to the worst performing 20% and they improve the following year, it does not prove that the program worked. The table above shows an “improvement” of $-1.5269 \rightarrow -0.7200$ even without any program. This is the strongest evidence for the idea of a control group in week 1.

Extrapolation: when does the model give a meaningless answer



year	$\hat{y}$	status
2014	17.8894	data
2018	20.2416	data
2025	24.3578	data
2050	39.0586	extrapolation
2100	68.4603	extrapolation
		meaningless

The model does not warn

The model says 68.46 t/ha for 2100. Today's world record is $\approx 45\text{--}50$ t/ha.

The model gave a meaningless answer and did not give any sign — because it is just a straight line. $R^2 = 0.869162$ does not show this either.

Rule: the model is valid only in the interval where the data are observed. Here $x \in [0; 8]$.

The second type of extrapolation

The model was built for Kazakh potatoes. If we apply it to Uzbekistan: RMSE = 12.514502 t/ha, and the model's own R^2 is 0.667950.

That is, 18.7357 times worse. The x interval was correct — the object was wrong.

**Summar
y**

Common student errors and misconceptions

1-5 1. “ r^2 is high $\Rightarrow$ the model is correct.” In all four of Anscombe’s equations, $r^2 = 0.666$; only one is valid.

2. “ r^2 is low $\Rightarrow$ the model is bad.” In Kyrgyzstan, $r^2 = 0.0029$ is the result: no trend.

3. Confusing r^2 and r . r^2 is unitless and does not give prediction accuracy. Accuracy is r .

4. Always interpreting r_0 . Unless $r = 0$ is in the data range, r_0 is just part of the line.

5. “The slope shows cause and effect.” No. That’s the whole topic of Week 13.

6-10 6. Consider $r_1 \sim r_2$ and $r_1 \sim r_2$ to be the same. $r_1 \cdot r_2 = 1$, not equal. Two different lines.

7. Call large leverage “influential”. The largest leverage in Tajikistan is in 1992/2018, and the influential point is 1995.

8. Automatically delete outliers. The decision comes from the origin of the data, not from the result.

9. Test for normality, do not check for linearity. The most important condition is the easiest to see — see the diagram.

10. Extrapolation. $\hat{r}(2100) = 68.46$ t/ha. The model does not warn — you warn.

Summary: three sentences

Regression analysis procedure

1 Scatter plot. Is it linear? Are there outliers? 2 Construct the line: β_1 , β_0 . 3 Interpret β_1 in terms of units. 4 Bring β_2 and β_1 together. 5 Residual plot: L and E. 6 Residuals by order: I. QQ: N. 7 Cook's distance: is there an influential point? 8 Prediction — only in the data range. 9 What is not proven — say.

In three sentences

1. Constructing the line is the easiest part. The formulas always give a number, even if the data is invalid.
2. The value of the model is in the residuals, not in β_2 . They tell where the model doesn't work — and sometimes they find errors in the data (31 movies).
3. Every model is limited by an interval and an object. A line doesn't know this limit.

Next step

Today we calculated $\beta_1 = 0.5880$. But this sample estimate — if we take another 9 years, it will give a different number.

In week 13: standard error for β_1 , t -test and confidence interval; prediction interval; and finally — correlation and causality, Simpson's paradox, confounding.

References

Main reading

►Cetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics (IMS, 2nd ed.), Chapter 7 (linear regression) and Chapter 24. Open source: openintro.org/book/ims ►Newbold P., Carlson W., Thorne B. Statistics for Business and Economics (9th ed.), 11.1–11.5. ►Heumann C., Schomaker M., Shalabh. Introduction to Statistics and Data Analysis, Chapter 11.

This week's required reading — both short

►Anscombe F. (1973). Graphs in statistical analysis. The American Statistician 27(1), 17–21. — 5 pages, and the most cited figure in the history of statistics. ►Galton F. (1886). Regression towards mediocrity in hereditary stature. Journal of the Anthropological Institute 15, 246–263. — This is where the word “regression” comes from; the first 6 pages are sufficient.

Supplement

►Cook R. D. (1977). Detection of influential observation in linear regression. Technometrics 19(1), 15–18. ►Durbin J., Watson G. (1951). Testing for serial correlation in least squares regression II. Biometrika 38(1/2), 159–177. ►Kahneman D. (2011). Thinking, Fast and Slow, Chapter 17 (“Regression to the mean”) — an example of praising and criticizing pilots.